Model Predictive Control
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Solution 06
Feasibility and Stability

Exercise 1 MPC and Feasible Set

1. We can set $P_N = 0_{2\times 2}$ since the terminal set is the point $[0,\ 0]^T$. If the control problem is feasible the terminal state must be zero, so any different matrix will make no difference.

2. The following code is used to transform the constrained finite time optimal control problem to a QP with substitution:

```
% system matrices
A = [1 0.1; 0 0.5]; % state matrix
B = [0;1];         % input matrix
N = 5;             % horizon
Q = eye(2);        % state cost
P = zeros(2);      % final state cost
R = 0.1;           % input cost
n = size(A,1);     % state dimension

% cost function matrices
QQ = blkdiag(kron(eye(N),Q),P);
RR = kron(eye(N),R);

% polytopic constraints
A_x = [1 0; 0 1; -1 0; 0 -1];
b_x = [15; 15; 15; 15];
A_u = [1; -1];
b_u = [1; 1];
A_f = [1 0; 0 1; -1 0; 0 -1];
b_f = [0; 0; 0; 0];

% constructing the QP with substitution
S_x = [];
S_u = [zeros(n,N)];
G = [kron(eye(N), A_u); A_x * S_u];
E = [zeros(2*N,n)];
for i = 0:N-1
    S_x = [S_x; A^i];
    E = [E; -A_x * A^i];
    tmp = [];
    for i = 0:i
        tmp = [A^i*B tmp];
    end
    tmp = [tmp zeros(n,N-i-1)];
    S_u = [S_u; tmp];
    if i < N-1
        G = [G; A_x * tmp];
    else
        G = [G; A_f * tmp];
    end
end
S_x = [S_x; A^N];
E = [E; -A_f * A^N];
```

```
w = [repmat(b_u,N,1); repmat(b_x,N,1); b_f];

% cost function terms for quadprog
H = (S_u'*QQ*S_u+RR);
F = (S_x'*QQ*S_u);
```

The code below computes the optimal input $u_0^*(x_0)$ at time step 0 at the desired grid points:

```
% initial conditions
x1 = -15:1:15;
x2 = x1;

% number of points in one axis of the grid
N_grid = size(x1,2);

% optimal input matrix
u_0_opt = zeros(N_grid, N_grid);

% testing the MPC for all the initial conditiojns
for i=1:N_grid
    for j=1:N_grid

        % setting the initial condition
        x0 = [x1(i);x2(j)];

        % the linear term depends on the initial condition
        f = F'*x0;

        % solving the problem
        [sol, ~, exitflag] = quadprog(H,f,G,w + E*x0,[],[],[],[]);

        % if the solver succeded then store the solution
        if exitflag==1
            u_0_opt(i,j) = sol(1);
        else
            u_0_opt(i,j) = NaN;
        end
    end
end

% matrices for scatter plot
X = [];
Y = [];
Z = [];
for i=1:N_grid
    for j=1:N_grid
        if ~isnan(u_0_opt(i,j))
            X = [X; x1(i)];
            Y = [Y; x2(j)];
            Z = [Z; u_0_opt(i,j)]
        end
    end
end

% using scatter to plot the region of attraction
scatter(X(:), Y(:),25,Z,'filled');
axis([min(x1) max(x1) min(x2) max(x2)])
xlabel('x1')
ylabel('x2')
```

Figure 1 shows the optimal $u^*(x_0)$ found at each point of the 30×30 grid.

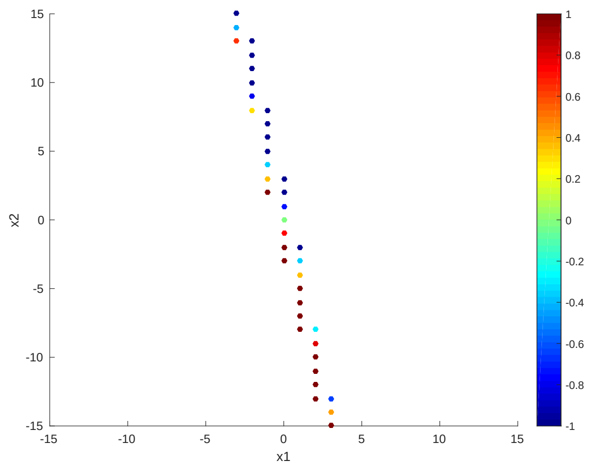

Figure 1: Feasible set for $N = 5$

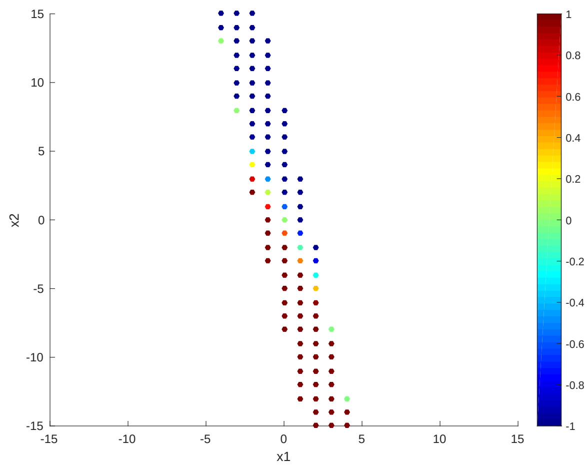

Figure 2: Feasible set for $N = 10$

3. If we increase the horizon to $N = 10$ we enlarge the feasible set, as we can see in Figure 2.

Exercise 2 **Convexity and stability**

1. If $x \in \alpha X_1 \oplus (1-\alpha)X_2$, then there exists an $x_1 \in X_1$ and $x_2 \in X_2$ such that $x = \alpha x_1 + (1-\alpha)x_2$. From linearity, we have that $x(k+1) = \alpha A x_1(k) + (1-\alpha)Ax_2(k)$. Invariance of X_1 and X_2 imply that $Ax_1 \in X_1$ and $Ax_2 \in X_2$ which in turn implies $Ax \in \alpha X_1 \oplus (1-\alpha)X_2$.

2. If $x \in \alpha X_1 \oplus (1-\alpha)X_2$, then there exists $x_1 \in X_1$ and $x_2 \in X_2$ such that $x = \alpha x_1 + (1-\alpha)x_2$. Convexity then gives the result.

3.

$$
\begin{aligned}
V(x(k+1)) - V(x(k)) &= \alpha x^T A^T P_1 A x + (1-\alpha)x^T A^T P_2 A x - \alpha x^T P_1 x - (1-\alpha)x^T P_2 x \\
&\leq -\alpha x^T \Gamma x - (1-\alpha)x^T \Gamma x \\
&= -x^T \Gamma x
\end{aligned}
$$

4.
   - The stage cost is a positive definite function
     Q and R are positive definite.
   - The terminal set is invariant under the local control law $\kappa_f(x) = Kx$
     Part 1) proved this
   - All state and input constraints are satisfied in $\mathcal{X}_f$
     Part 2) proved that the state constraints are satisfied. The proof for the input constraints is the following:

$$
\begin{aligned}
x_N &\in \alpha X_1 \oplus (1-\alpha)X_2 \\
Kx_N &\in K(\alpha X_1 \oplus (1-\alpha)X_2) \\
Kx_N &\in \alpha K X_1 \oplus (1-\alpha)KX_2 \\
Kx_N &\in \alpha \mathcal{U} \oplus (1-\alpha)\mathcal{U} \\
Kx_N &\in \mathcal{U}
\end{aligned}
$$

   - The terminal cost is a Lyapunov function in the terminal set $\mathcal{X}_f$
     Part 3) proved this

Exercise 3 **Invariant Terminal Set**

1. The input sequence to apply is

$$
\begin{aligned}
u(1) &= u_1^\star \\
u(2) &= u_2^\star \\
u(3) &= u_3^\star \\
u(4) &= Kx(4) \\
u(5) &= Kx(5)
\end{aligned}
$$

2. The first reason is that re-optimization provides robustness to any noise or modeling errors, while the second is that the solution at time $k = 0$ is sub-optimal because it is over a finite horizon. Re-optimizing can provide a control law with better performance.

Exercise 4 **Stability and Recursive Feasibility of Nominal MPC**

i) Find the infinite horizon LQR controller by solving the Algebraic Riccati Equation:

$$
\begin{aligned}
p &= a^2 p - \frac{a^2 b^2 p^2}{r + b^2 p} + q \\
0 &= (a^2 - 1)(r + b^2 p)p - a^2 b^2 p^2 + q(r + b^2 p) \\
0 &= -p^2 b^2 + (ra^2 - rp + qb^2)p + qr
\end{aligned}
$$

with $a = 2$,$b = 1$, $r = q = 1$, resulting in

$$0 = p^2 - 4p - 1,$$

with the solutions

$$p = 2 \pm \sqrt{5}.$$

We take the unique positive definite solution to obtain

$$K_\infty = -\frac{abp}{r + b^2 p} = -\frac{4 + 2\sqrt{5}}{3 + \sqrt{5}}$$

ii) For any asmpt. stable scalar linear system the set $[-1, 1]$ is invariant, and the LQR stabilizes the system. Alternatively, we can check that $|a + bk| < 1$ is satisfied.
To check constraint satisfaction we need to consider $|Kx_{max}| = |K| = |\frac{4+2\sqrt{5}}{3+\sqrt{5}}| = |2\frac{2+\sqrt{5}}{3+\sqrt{5}}| < 2$ so $|u| = |Kx| \leq 2$.

iii) The terminal set is invariant under a local control law while satisfying the constraints. Therefore, if we have a feasible solution to the MPC problem at time step zero 0 with states $\{x_0^*, x_1^*, \ldots, x_N^*\}$ and inputs $\{u_0^*, u_1^*, \ldots, u_{N-1}^*\}$, at the next time step 1 we can construct a feasible solution as

$$\{x_1^*, x_2^*, \ldots, (a + bK_\infty)x_N^*\} \text{ and } \{u_1^*, u_2^*, \ldots, K_\infty x_N^*\}$$

which shows recursive feasibility.

iv)
- Since the problem is recursively feasible and feasible for the initial state, the state of the closed-loop system will always stay within the feasible set. The feasible set is bounded and contains the origin, thus the system state will stay within a bounded set.
- There is no terminal cost and therefore the necessary cost decrease for asymptotic stability is not ensured.
- Adding a terminal cost based on the solution of the infinite horizon LQR control problem can ensure asymptotic stability.